

SOAL 1

```
1) for (int i = n; i >= 1; i /= 2)
2)     printf("%d", n);
```

command	time	iteration
---------	------	-----------

1	c1	k + 1
2	c2	k

$T(n) = c1 (k + 1) + c2 (k) = (c1 + c2) k + c1 = (c3)k + c1$

$n / (2 * 2 * 2 * \dots) = 1$

$n / (2 ** (k - 1)) = 1 \rightarrow n = 2 ** (k - 1) \rightarrow \log_2 n = k - 1 \rightarrow k = \log_2 n + 1$

$T(n) = (c3)(\log_2 n + 1) + c1$

$T(n) = (c3)(\log_2 n) + c4$

SOAL 2

```
1) for (int i = 1, j = n; i <= j; i *= 2, j /= 2)
2)     printf("%d", i + j);
```

command	time	iteration
---------	------	-----------

1	c1	k + 1
2	c2	k

$T(n) = c1 (k + 1) + c2 (k) = (c3) k + c1$

$n / (2 * 2 * \dots) = 1 * (2 * 2 * \dots)$

$n / (2 ** (k - 1)) = 1 * (2 ** (k - 1))$

$n = (2 ** (k - 1)) * (2 ** (k - 1))$

$n = (2 ** (2k - 2))$

$\log_2 n = 2k - 2 \rightarrow k = (\log_2 n + 2) / 2$

$T(n) = (c3) ((\log_2 n + 2) / 2) + c1$

$T(n) = (c3 (\log_2 n) + 2 c3) / 2 + c1$

$T(n) = (c3 (\log_2 n) / 2) + c3 + c1$

$T(n) = (c4) (\log_2 n) + c5$

SOAL 3

```
1) for (int i = 1, j = n; i <= j; i += 2, j -= 2)
2)     printf("%d", i + j);
```

command	time	iteration
---------	------	-----------

1	c1	k + 1
2	c2	k

$T(n) = c1 (k + 1) + c2 (k) = (c3) k + c1$

$n - (2 + 2 + \dots) = 1 + (2 + 2 + \dots)$

$n - 2(k - 1) = 1 + 2(k - 1)$

$n = 1 + 4(k - 1) = 1 + 4k - 4 = 4k - 3$

$(n + 3) / 4 = k$

$T(n) = (c3) ((n + 3) / 4) + c1$

$T(n) = (c3) (n / 4 + 3 / 4) + c1$

$T(n) = (c4) (n) + c5$

SOAL 7

```
1) for (int i = 1; i <= n - 1; i++)
2)     for (int j = 2; j <= n; j++)
3)         for (int p = 3; p <= n - 2; p++)
4)             printf("%d", i + j + p);
```

command	time	iteration
---------	------	-----------

1		c1		n
2		c2		(n - 1) * (n)
3		c3		(n - 1) * (n - 1) * (n - 3)
4		c4		(n - 1) * (n - 1) * (n - 4)

$T(n) = (c1)(n) + (c2)((n - 1)(n)) + (c3)((n - 1)(n - 1)(n - 3)) + (c4)((n - 1)(n - 1)(n - 4))$
 $T(n) = (c1)(n) + (c2)(n^2 - n) + (c3)(n^3 - 5n^2 + 7n - 3) + (c4)(n^3 - 6n^2 + 9n - 4)$
 $T(n) = (c5)(n) + (c6)(n^2) + (c7)(n^3) + c8$

SOAL 8

```

1) for (int i = 1; i <= n; i++)
2)     for (int j = 1; j <= n; j++)
3)         for (int p = 1; p <= i + j; p++)
4)             printf("%d", i + j + p);

```

command	time	iteration
1		c1 n + 1
2		c2 (n) * (n + 1)
3		c3 $\sum_{i=1}^n \sum_{j=1}^n (i + j + 1)$
4		c4 $\sum_{i=1}^n \sum_{j=1}^n (i + j)$

$T(n) = (c1)(n + 1) + (c2)(n * (n + 1)) + (c3)(\sum_{i=1}^n \sum_{j=1}^n (i + j + 1)) + (c4)(\sum_{i=1}^n \sum_{j=1}^n (i + j))$

$\sum_{i=1}^n \sum_{j=1}^n (i + j + 1) = (n(n(n + 1) / 2)) + (n * (n + 1) / 2) + n^2 = (n^3 + n^2) / 2 + (n^2 + n) / 2 + n^2 = 1/2(n^3) + 1/2(n^2) + 1/2(n^2) + 1/2(n) + n^2 = 1/2(n^3) + (1/2 + 1/2 + 1)(n^2) + 1/2n = 1/2(n^3) + 2n^2 + 1/2n = (1/2n)(n^2 + 4n + 1) = (n)(n^2 + 4n + 1) / 2 = (n^3 + 4n^2 + n) / 2$

$\sum_{i=1}^n \sum_{j=1}^n (i + j) = (n(n(n + 1) / 2)) + (n * (n + 1) / 2) = (n^3 + n^2) / 2 + (n^2 + n) / 2 = 1/2(n^3) + 1/2(n^2) + 1/2(n^2) + 1/2(n) = 1/2(n^3) + (1/2 + 1/2)(n^2) + 1/2n = 1/2(n^3) + n^2 + 1/2n = (1/2n)(n^2 + 2n + 1) = (n^3 + 2n^2 + n) / 2$

$T(n) = (c1)(n) + (c1) + (c2)(n^2) + (c2)(n) + (c3)(n^3 + 4n^2 + n) / 2 + (c4)(n^3 + 2n^2 + n) / 2$
 $T(n) = (c1)(n) + (c1) + (c2)(n^2) + (c2)(n) + (c3/2)(n^3) + (c3)(2n^2) + (c3/2)n + (c4/2)(n^3) + (c4)(n^2) + (c4/2)(n)$
 $T(n) = (n)(c1 + c2 + c3/2 + c4/2) + (n^2)(c2 + 2c3 + c4) + (n^3)(c3/2 + c4/2) + c1$
 $T(n) = (c5)(n) + (c6)(n^2) + (c7)(n^3) + c1$

SOAL 9

```

1) for (int i = 1; i <= n; i++)
2)     for (int j = 1; j <= n - i; j++)
3)         printf("%d", i + j);

```

command	time	iteration
1		c1 n + 1
2		c2 $\sum_{i=1}^n (n - i + 1)$
3		c3 $\sum_{i=1}^n (n - i)$

$T(n) = (c1)(n + 1) + (c2)(\sum_{i=1}^n (n - i + 1)) + (c3)(\sum_{i=1}^n (n - i))$

$\sum_{i=1}^n (n - i + 1) = n^2 - (n(n + 1) / 2) + n = n^2 - (n^2 + n) / 2 + n = n^2 - (n^2) / 2 - n / 2 + n = n^2 - (n^2) / 2 - n / 2 + 2n / 2 = n^2 - (n^2) / 2 + n / 2$
 $\sum_{i=1}^n (n - i) = n^2 - (n(n + 1) / 2) = n^2 - (n^2 + n) / 2 = n^2 - (n^2) / 2 - n / 2$

$T(n) = (c1)(n) + (c1) + (c2)(n^2) + (-c2/2)(n^2) + (-c2/2)(n) + (c2)(n) + (c3)(n^2) + (-c3/2)(n^2) + (-c3/2)(n)$
 $T(n) = (c1 - c2/2 + c2 - c3/2)(n) + (c2 - c2/2 + c3 - c3/2)(n^2) + c1$
 $T(n) = (c4)(n) + (c5)(n^2) + c1$